G.C.E. (O/L) · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 02

Mathematics I

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Time: 2 hours **Total marks:** 100

Index number: .....

Instructions:

- This paper has two parts, **Part A** and **Part B**. Answer **all** questions on this question paper itself.
- Use only the space provided under each question for your working and your answer.
- Show the relevant steps and write the correct units in your answers.
- Part A: 2 marks for each question. Part B: 10 marks for each question.
- Take $\pi = \frac{22}{7}$ where necessary. Scrap paper may be used for rough calculations.

Examiner's use only	
Part	**Marks**
Part A (Questions 1 – 25)	/ 50
Part B Question 1	/ 10
Question 2	/ 10
Question 3	/ 10
Question 4	/ 10
Question 5	/ 10
Total	**/ 100**

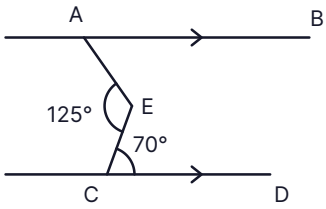
Part A

Answer all questions on this paper itself. Take $\pi = \frac{22}{7}$. (2 × 25 = 50 marks)

1. In a factory, 8 workers can complete an order in 18 days. How many more workers, working at the same rate, must be employed to complete the order in 12 days?

Answer: .....

2. In the figure, AB ∥ CD, $A\hat{E}C = 125°$ and $E\hat{C}D = 70°$. Find the size of $B\hat{A}E$.



Answer: .....

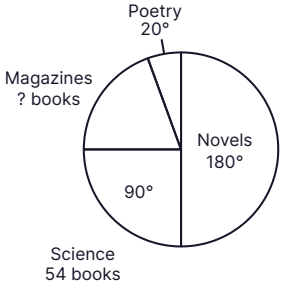
3. Simplify: $\frac{3}{4x} - \frac{5}{6x} + \frac{2}{3x}$

Answer: .....

4. Which one of the following **cannot** be a face of a right triangular prism? Write its number in the box.
(i) a square (ii) a rectangle whose adjacent sides are unequal (iii) a parallelogram which has no right angle

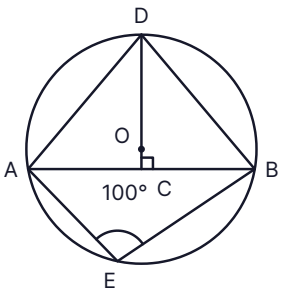
Answer: ( ) .....

5. The pie chart shows how the books borrowed from a library on a certain day were divided among four categories. 54 Science books were borrowed. Find the number of Magazines borrowed.



Answer: .....

6. In the figure, O is the centre of the circle and DOC is a straight line which meets the chord AB at its mid-point C. If $A\hat{E}B = 100°$, find the size of $C\hat{D}B$.



Answer: .....

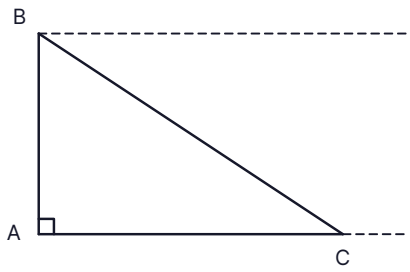
7. Factorise: $4x^2 - 36$

Answer: .....

8. Given that lg 3.58 = 0.5539, write down the value of lg 358.

Answer: .....

9. In the figure, AB is a vertical chimney of a factory and AC is level ground. A bird is at B, the top of the chimney, and a rat is at C. On the figure mark the angle of elevation of B from C as a and the angle of depression of C from B as b, and write the relation between a and b.



Answer: .....

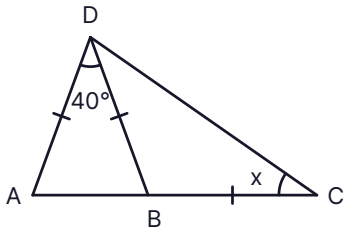
10. A factory makes cylindrical tins. The height of a tin is 5 times its base radius r. Write, in its simplest form, the ratio of the area of paper needed to cover the curved surface of a tin to the area of paper needed to cover one circular base. (Curved surface area of a cylinder = $2\pi rh$)

Answer: .....

11. Two fair tetrahedral dice have faces numbered 1, 2, 4, 5. Both dice are rolled together and the numbers on their bottom faces are added. Find the probability that the sum is 6.

Answer: .....

12. In the figure, ABC is a straight line and AD = DB = BC. If $A\hat{D}B = 40°$, find the value of x.



Answer: .....

13. The universal set is $\xi = \{x : x \text{ is an integer}, 1 \leq x \leq 12\}$. A = {multiples of 3} and B = {even numbers}, both taken from ξ. Write the number of the correct statement.
(i) $A \cap B = \{6\}$ (ii) $n(A \cup B) = 8$ (iii) $n(A) = 3$

Answer: ( ) .....

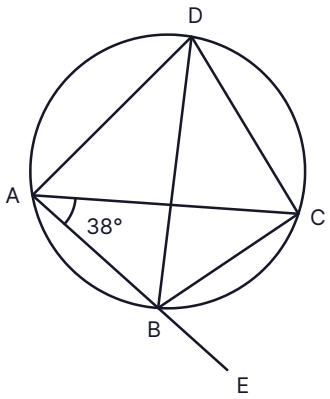
14. Simplify: $(6x^2y^3 \times 4x^3y) \div 12xy^2$

Answer: .....

15. In a geometric progression the second term is −2 and the third term is −6. Find the fifth term.

Answer: .....

16. ABCD is a cyclic quadrilateral. BD bisects AD̂C, AB is produced to E and BÂC = 38°. Find the size of CB̂E.



Answer: .....

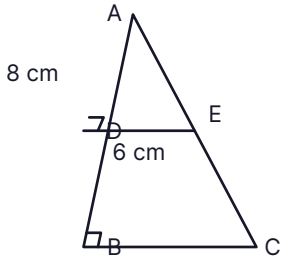
17. Find the highest common factor of $8a^3b^2$, $12a^2b^3$ and $20a^2b$.

Answer: .....

18. A solid metal prism with a triangular base of area 154 cm^2 is melted and recast, without loss, into a solid cylinder of the same height and of base radius r cm. Find r. (Volume of a cylinder = $\pi r^2 h$)

Answer: .....

19. In triangle ABC, D and E are the mid-points of AB and AC respectively. AD̂E = 90°, AD = 8 cm and DE = 6 cm. Find the perimeter of the quadrilateral BCED.

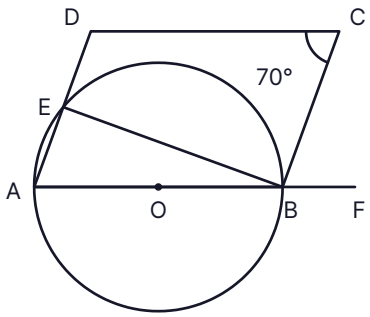


Answer: .....

20. Find the equation of the straight line passing through the points (2, 1) and (5, 7).

Answer: .....

21. In the figure, O is the centre of the circle and AOBF is a straight line. ABCD is a parallelogram and E is the point where AD meets the circle. If $B\hat{C}D = 70°$, find the size of $E\hat{B}F$.

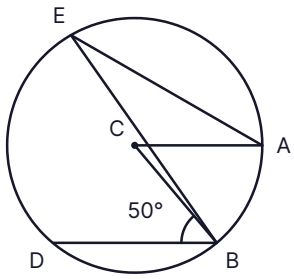


Answer: .....

22. Pradeep invested a sum of money for 2 years at an annual compound interest rate of 8%. The interest he earned in the second year was Rs. 2 160. Find the amount he invested.

Answer: .....

23. In the figure, C is the centre of the circle and AC ∥ BD. If $C\hat{B}D = 50°$, find the size of $A\hat{E}B$.

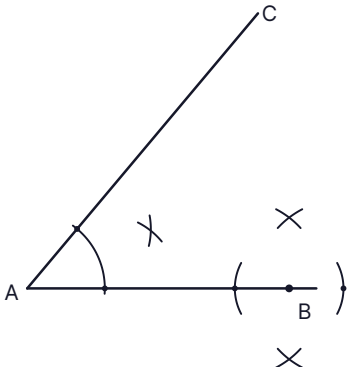


Answer: .....

24. Solve the inequality $2x + 9 > 1$ and write the negative integers which satisfy it.

Answer: .....

25. AB and AC are two tangents drawn from A to a circle, and AB touches the circle at B. Some of the construction lines needed to find the centre O of the circle are shown. Using them, complete the construction and mark the centre O. (The construction lines must be clearly visible.)



Part B

Answer all questions on this paper itself. (10 × 5 = 50 marks)

1. Tharindu works in a factory. He pays income tax at 10% on the part of his monthly salary that exceeds Rs. 100 000. After paying the tax, he spends $\frac{1}{4}$ of the money he receives on food, and $\frac{2}{5}$ of the money that is then left on other expenses.

(i) What fraction of the money he receives is left after spending on food? (1 mark)

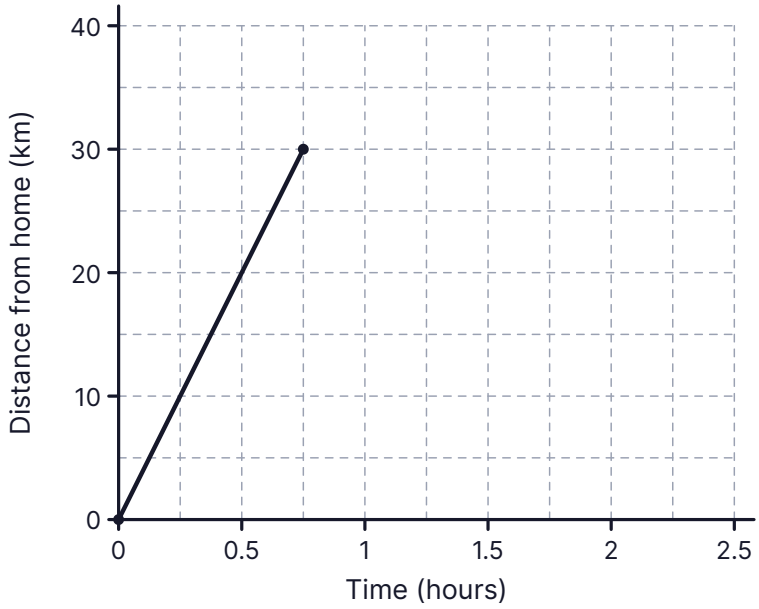
(ii) What fraction of the money he receives is left after spending on food and on other expenses? (2 marks)

(iii) The amount left after these two expenses is Rs. 55 125. Find the amount spent on food and the amount spent on other expenses separately. (2 marks)

(iv) Find Tharindu's monthly salary before tax. (3 marks)

(v) Express the amount spent on food as a percentage of his monthly salary before tax. (2 marks)

2. (a) Sanduni goes by bus from her home to the town where the library is. The bus travels for 45 minutes at a uniform speed and she covers 30 km; this is shown as the first part of the distance-time graph below. She stays at the library for 30 minutes and then returns home by the same route in 1 hour, at another uniform speed.



(i) Complete the distance-time graph on the grid. (2 marks)

(ii) Find the speed of the bus on the return journey, in km h^{-1}. (1 mark)

(iii) Using the graph, find how long after she left home she was 15 km from home on the return journey. (1 mark)

(b) The figure shows a plane figure made of a semicircle AEB joined to a rectangle ABCD. AB = 14 cm, BC = 55 cm and $B\hat{C}D$ is a right angle. (Take $\pi = \frac{22}{7}$.)



(i) Find the perimeter of the figure. (3 marks)

(ii) The area of the rectangle is how many times the area of the semicircle? (3 marks)

3. Amaya has rented out a building to a library for Rs. 10 000 per month, and she receives the rent for the whole year in one payment. The annual assessed value of the building is Rs. 80 000, and the municipal council charges annual rates of 15% of the assessed value. She spends 10% of the rent received on maintenance. She then adds some more money to the amount left and buys shares of Rs. 80 each. Each share pays an annual dividend of Rs. 4, and she receives a total dividend of Rs. 6 000 for the year.

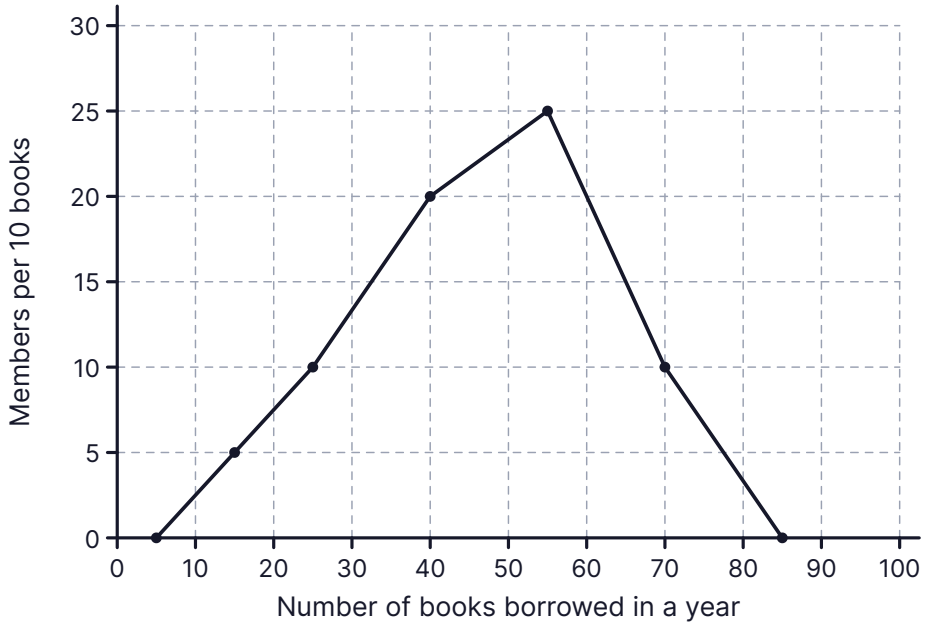
(i) Find the annual rates payable. (2 marks)

(ii) Find the amount left after paying the rates and the maintenance cost. (2 marks)

(iii) (a) Find the number of shares she bought and the total amount she invested in them. (3 marks)

(b) Express the extra money she added as a percentage of the total amount invested. (3 marks)

4. The members of a library borrowed books during a year. The frequency polygon below shows the distribution of the number of books borrowed. The vertical axis shows the number of members **per 10 books** in each class, and the points of the polygon are plotted at the mid-values of the classes. (In a class such as 10 – 20, 10 is included and 20 is not.) An incomplete frequency table is also given.



Number of books	Number of members	Cumulative frequency
10 – 20	5	5
20 – 30	.....	.....
30 – 50	.....	.....
50 – 60	.....	.....
60 – 80	.....	.....

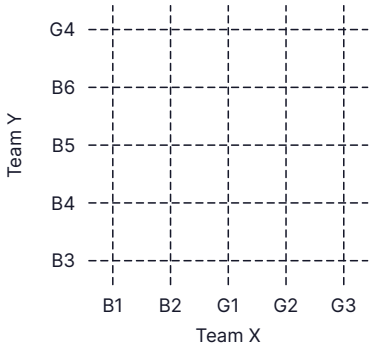
(i) Complete the frequency column of the table using the polygon. (3 marks)

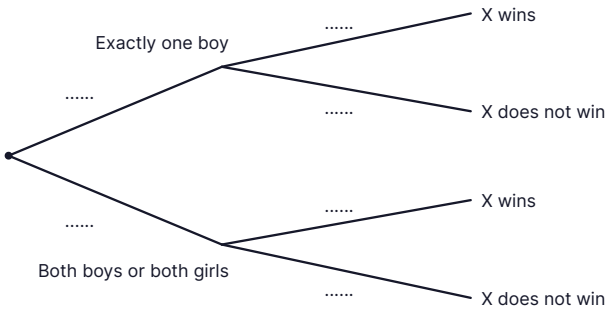
(ii) Draw the histogram corresponding to the table on the grid given. (3 marks)

(iii) Complete the cumulative frequency column. (2 marks)

(iv) If a member is chosen at random, find the probability that the member borrowed 50 or more books. (2 marks)

5. In a library club, Team X has 2 boys (B1, B2) and 3 girls (G1, G2, G3), and Team Y has 4 boys (B3, B4, B5, B6) and 1 girl (G4). One member is chosen at random from each team to represent the club in a quiz. The grid below is given to show the outcomes.





(i) Mark the sample space of the selection of the two members on the grid, using a cross (×) for each outcome. (2 marks)

(ii) Enclose with closed curves the outcomes in which exactly one boy is selected, and find the probability of that event. (3 marks)

(iii) If exactly one boy is selected, the probability that Team X wins is $\frac{3}{7}$; if both are boys or both are girls, the probability that Team X wins is $\frac{2}{5}$. Complete the probabilities on the tree diagram. (3 marks)

(iv) Find the probability that Team X does not win. (2 marks)

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